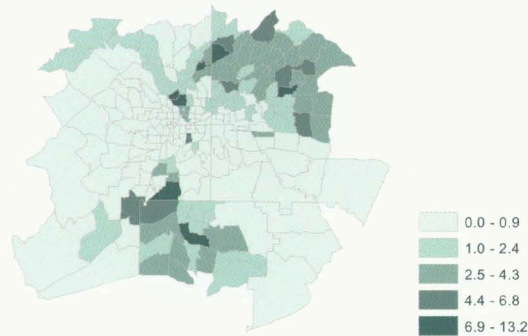


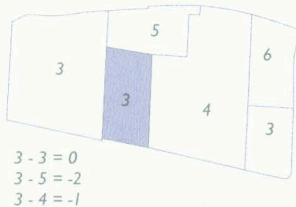
MEASURING THE SIMILARITY OF NEARBY FEATURES

Geary's c and Moran's I use the magnitude of feature values to identify and measure the strength of spatial patterns. A social scientist studying the integration of different ethnic groups in a region could measure the extent to which the groups are clustered or dispersed, using the percentage of the population of each group in each census tract or block group.

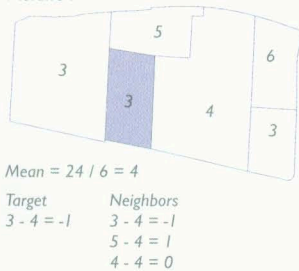


Census tracts color coded by percent Asian population.

Geary's c



Moran's I



What Geary's c and Moran's I measure

Both methods compare values for neighboring features. They then compare the difference in values between each pair of neighbors to the difference in values between all features in the study area. If the average difference between neighboring features is less than between all features, the values are clustered.

Geary's c calculates the difference in values between the target features and each of its neighbors. Moran's I calculates the difference between the target feature and the mean for all features, and the difference between each neighbor and the mean. It then compares these differences for the target feature and each neighbor in turn.

For both Geary's c and Moran's I , you can specify a neighborhood based on adjacency, a set distance, or the distance to all features in the dataset. Choose according to the nature of the spatial relationship between the features you're analyzing. Defining a neighborhood is discussed in "Defining spatial neighborhoods and weights."

Calculating Geary's c

The GIS starts with one feature and subtracts the value of a neighboring feature from the value of the original, or target, feature.

Since the GIS only needs to find out how large the difference is between neighboring features, and not whether the difference is positive or negative, it squares the difference to make sure it's positive.

The GIS then multiplies the difference by a weight value (based on the neighborhood you defined). It repeats the process for the target feature with all other features in the study area. It then moves to the next feature and does the same thing, summing the results as it goes.

For each target (i) and neighbor (j) pair, the value of the neighbor (x_j) is subtracted from the value of the target (x_i), the difference is squared, and the result multiplied by the weight (w_{ij}) for that pair....

$$\sum_i \sum_j w_{ij} (x_i - x_j)^2$$

....then the results are added to the sum for all pairs

Next the GIS calculates the variance—how far values are from the mean. (See “Understanding data distributions.”)

Now the GIS sums the weights for each pair of features, multiplies this by two to account for each pair-wise combination (to-from and from-to), and then multiplies the variance by this result. It does this to account for the weights that were summed when calculating the difference in feature values.

The weights for all pairs are summed, and the result doubled....

$$2 \sum_i \sum_j w_{ij} \frac{\sum_i (x_i - \bar{x})^2}{n}$$

....then multiplied by the variance (the difference between each target value and the mean value squared, the result summed for all features, and divided by the number of features)

Finally, it divides this value into the initial value it calculated (the sum of the weighted differences in values) to get the c ratio. The number of features (n) is moved to the numerator to simplify the equation.

$$c = \frac{n \sum_i \sum_j w_{ij} (x_i - x_j)^2}{2 \sum_i \sum_j w_{ij} \sum_i (x_i - \bar{x})^2}$$

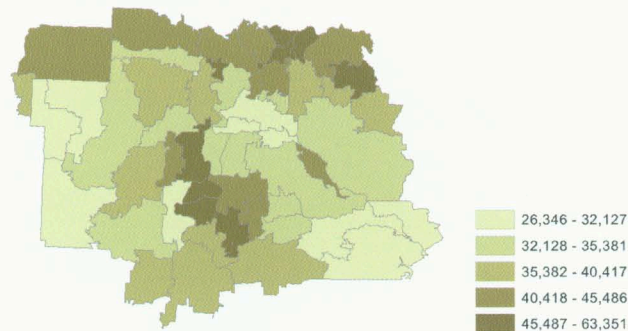
Since the weight value is used in both the numerator and denominator, the ratio is essentially the sum of the differences in attribute values between any two features within the neighborhood (as defined by the weights) compared to the sum of the differences for the dataset as a whole (as represented by the variance). Multiplying the differences by the weights incorporates the proximity of the features.

Interpreting Geary's c

A small difference in values between nearby features will result in a small numerator, indicating that similar values are more likely to be found near each other. Conversely, a large difference in values indicates that similar values are less likely to be found near each other.

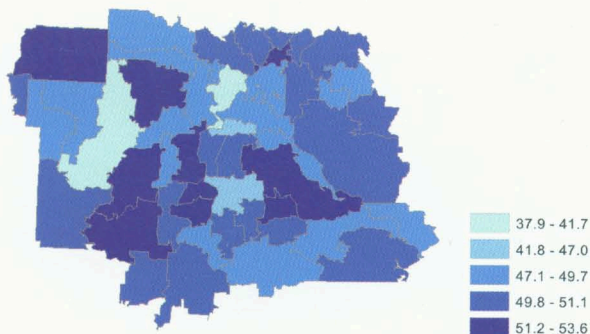
While that gives you a general sense of the pattern, you can also figure out how much the value of c differs from the value you would expect for a random distribution. A value of c equal to 1 means the distribution is random (the numerator and denominator in the equation are equal).

A value of c close to 0 means the distribution of values is clustered. If all the features in the study area had close to the same value, the sum of all the pairs would be near 0, the numerator would be a very small number, and the final value for c would also be close to 0.



ZIP Codes color coded by median income. A Geary's c value less than 1 (0.46 in this case) indicates clustering.

Conversely, a value of c greater than 1 indicates the values are dispersed. The numerator is larger than the denominator—the differences in values for neighboring features are greater than the difference between values for the study area as a whole.



ZIP Codes color coded by percent female population. A c value greater than 1 (1.26) indicates a dispersed pattern.

Calculating Moran's I

For each pair of features, the GIS subtracts the value of each feature from the mean value for all the features in the study area. It then multiplies these values to get what's known as the cross-product. The GIS repeats the process for all pairs of features in the study area and sums the results.

The mean is subtracted from the value of the target feature and the value of the neighbor, and the differences are multiplied...

$$\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})$$

...then the result is multiplied by the weight for that pair, and added to the sum for all features

The software then calculates the variance from the mean value for all the features in the study area, sums the weights for each pair of features, and multiplies the variance by this sum, just as with Geary's c .

The variance is multiplied by the sum of the weights

$$\sum_i \sum_j w_{ij} \frac{\sum_i (x_i - \bar{x})^2}{n}$$

Finally, it divides this value into the initial value it calculated (the sum of the weighted cross-products) to get the ratio. Again, like Geary's c , the number of features variable (n) is moved to the numerator.

Moran's Index

$$I = \frac{n \sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i \sum_j w_{ij} \sum_i (x_i - \bar{x})^2}$$

Interpreting Moran's *I*

In the equation for Moran's *I*, the GIS multiplies the difference from the mean for each pair of features—a high cross-product indicates nearby features have similar values; a low cross-product indicates nearby features have dissimilar values.

For example, if the mean is 6 and the target feature has a value of 12, the cross-product for a neighboring feature with a value of 10 is 24:

$$(12-6) \times (10-6) = 6 \times 4 = 24$$

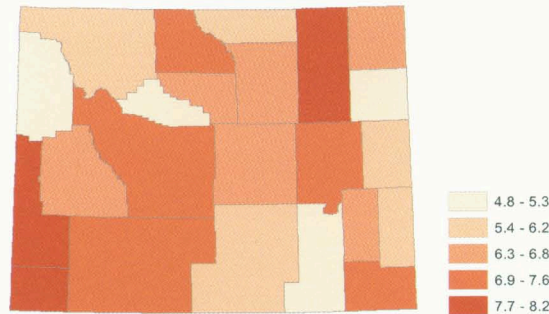
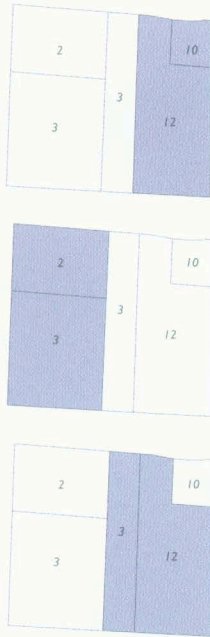
If both neighboring values are less than the mean—say 3 and 2—the cross-product is still positive, but is less than the cross-product for the higher values:

$$(3-6) \times (2-6) = -3 \times -4 = 12$$

If one value of the pair is greater than the mean and one is less than the mean, then the product is negative. For example, if the target feature has a value of 12 but the neighboring feature has a value of 3, the cross-product is negative, in this case, -18:

$$(12-6) \times (3-6) = 6 \times -3 = -18$$

If there are roughly as many pairs with positive cross-products as there are with negative cross-products, the result of summing the cross-products will be close to 0. That indicates a random distribution—some neighbors have similar values and some don't. So if you get a value for *I* that's close to 0, the distribution of feature values resembles a random distribution.



Percent of population 5 to 10 years of age, by county. An *I* of -0.06 (close to 0) indicates the distribution is similar to a random distribution.

You can calculate the expected value of *I* if the values are distributed randomly, using the number of features in the study area.

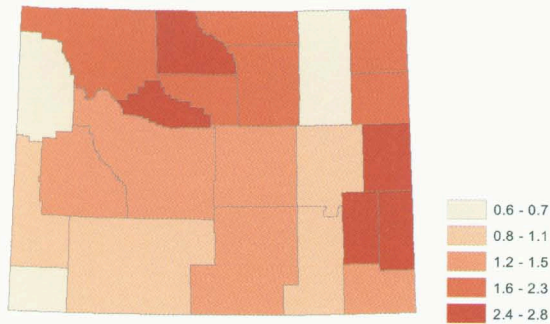
The expected Moran's *I* for a random distribution

$$I_E = \frac{-1}{n-1}$$

-1 is divided by the number of features (n), minus 1

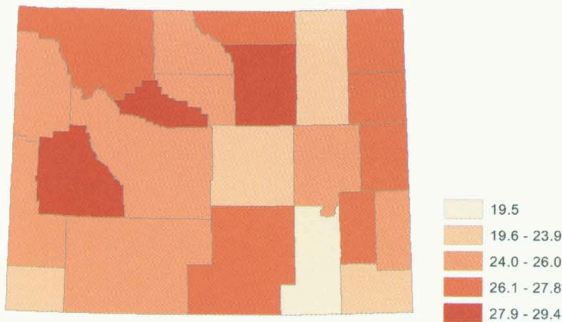
This turns out to be a small negative number, close to 0. So, for all practical purposes, a value of 0 for I is used to indicate a random distribution.

If more pairs have similar values than not, the sum of the cross-products will be positive, and I will be greater than 0. That means similar values are clustered.



Percent of population 85 years of age and above. An I greater than 0 (0.26, in this case) indicates a clustered pattern.

If there are more pairs with dissimilar values, the sum of the cross-products will be negative, and I will be less than 0. That means the values are dispersed.



Percent of population 45 to 65 years of age. An I of -0.12 indicates a slightly dispersed pattern since it's a little less than 0.

The range of possible values of I is -1 to 1 . If all neighboring features had close to the same value, the equation would calculate to near 1 , indicating complete clustering of values. Conversely, if the values are completely dispersed, the value of I calculates out to a value near -1 .

This table, based on one by Michael Goodchild, shows the possible values of Geary's *c* and Moran's *I*, and the pattern they indicate.

Value of Geary's <i>c</i>	Value of Moran's <i>I</i>	Pattern
<i>c</i> < 1	<i>I</i> > 0	Clustered (similar values are found together)
<i>c</i> = 1	<i>I</i> = 0	Random (no apparent pattern)
<i>c</i> > 1	<i>I</i> < 0	Dispersed (high and low values are interspersed)

Testing the significance of the results for Geary's *c* and Moran's *I*

For both Geary's *c* and Moran's *I*, the GIS calculates a Z-score to indicate how confident you can be that any pattern is not simply due to chance.

The GIS calculates the expected statistic for a random distribution, calculates the variance for that statistic, subtracts the expected statistic from the observed statistic, and divides by the square root of the variance (the standard deviation).

The Z-score for Geary's *c*

$$Z_c = \frac{c_o - c_e}{SD_{c_e}}$$

...and the difference divided by the standard deviation of *c* for the expected distribution

The expected *c* is subtracted from the observed *c*...

$$Z_I = \frac{I_o - I_e}{SD_{I_e}}$$

The Z-score for Moran's *I* is calculated the same way

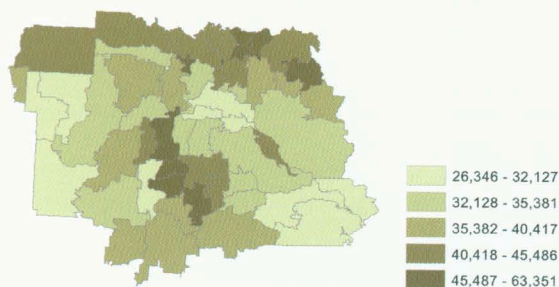
While the expected value of Geary's *c* for a random distribution is always 1, the expected value of Moran's *I* for a random distribution depends on the number of features in the study area, calculated as $-1/(n-1)$, where *n* is the number of features. This is a very small number, close to 0.

The calculation of the variance is based on the number of features in the study area, the number of neighboring features, and the sum of the weight values. The equation for the variance (and thus the Z-score value) depends on the sampling assumption you use—either randomization or normalization.

Some software asks you to specify the sampling assumption, while other software calculates the Z-score using both assumptions, or simply chooses one (in the latter case you should check the software documentation to see which assumption is used). Sampling is discussed in the section “Testing statistical significance.”

For Geary’s c , a positive Z-score indicates a dispersed pattern, while a negative Z-score indicates clustering. Since 1 (the expected value for a random distribution) is subtracted from the observed c to calculate the numerator of the equation, a c value less than 1 (indicating clustering) will result in a negative number for the numerator, and thus a negative value for Z . Conversely, a positive c value (indicating a dispersed pattern) will result in a positive number for the numerator, and a positive value for Z .

For Moran’s I , the opposite is true. Since the expected I is close to 0, when it is subtracted from the observed I , an observed I greater than 0 (indicating clustering) will result in a positive numerator, and a positive Z-score. An observed I less than 0 (indicating a dispersed distribution) will result in a negative numerator and a negative Z-score. So, a positive Z-score indicates clustering, while a negative Z-score indicates a dispersed pattern. See “Testing statistical significance” for more about the Z-score.

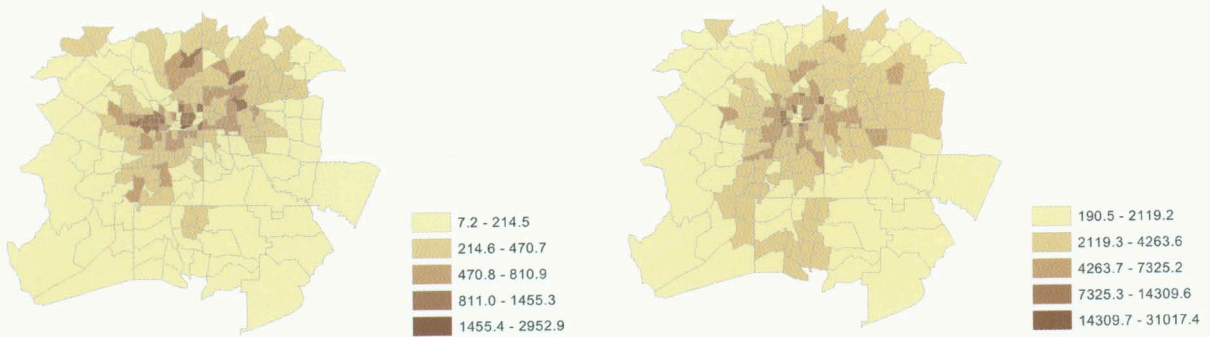


Median income by ZIP Code. Both Moran’s I and Geary’s c indicate clustering. The Z-score for Moran’s I is 4.34, and for Geary’s c is -4.95 . Those values exceed the critical value of 2.58 at a confidence level of 0.01 (-2.58 for Geary’s c), so using either measure you can be 99% sure the clustering didn’t occur by chance.

Using Geary’s c or Moran’s I to compare distributions

If you find there is a statistically significant pattern, you may want to see if there is an obvious reason for the pattern. For example, if you’re analyzing the distribution of senior citizens by ZIP Code, and find that they tend to live in clusters, you may also want to analyze the distribution of the population as a whole. It may be that the senior population is clustered simply because the population in general occurs in clusters across the region. You’d do this by analyzing the population by ZIP Code, using the same statistic, and calculating the Z-score. If the Z-score for the senior population is higher (using Moran’s I) than that for the population as a whole, you can conclude the senior population is more clustered than you

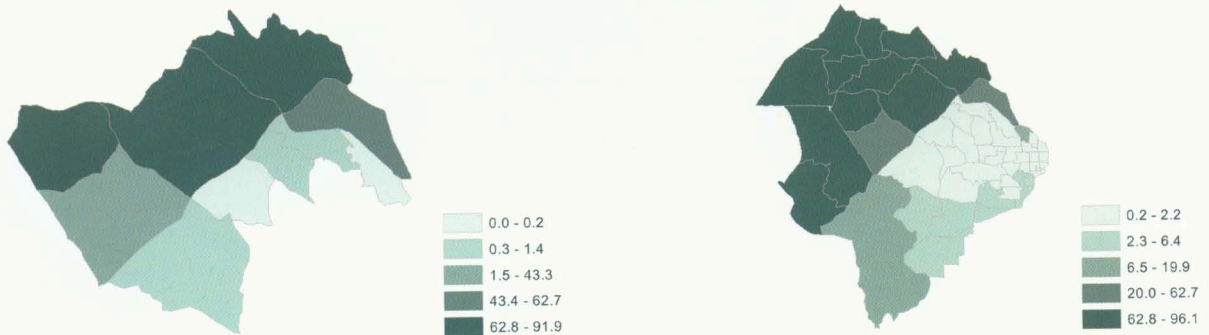
would expect based on the distribution of the general population. (See “Using statistics with geographic data.”)



These density maps (number of people per square mile) show that with an I of 0.3 and a Z-score of 8.14, the population 65 years of age and above (left) is slightly more clustered than the population in general (right), with an I of 0.26 and a Z-score of 4.39.

Factors influencing the results of Geary's c and Moran's I

The results of Geary's c and Moran's I are affected by geographic scale and the extent of the study area. A study area may have few features, the ones with similar values occurring near each other. However, since there are few features, areas with dissimilar values could also be within the neighborhood, or even adjacent, so the statistic will not show a clustered pattern. If you expanded the study area to include more features, the statistic would show the clustering of values.



Two study areas showing percent white population by census tract. Even though the values appear to be clustered in the map on the left, Moran's I indicates no pattern—because there are so few features, the features with high values are adjacent to features with low values. In the area on the right, Moran's I indicates very strong clustering, with an I of 0.76 and a Z-score of 9.5.

MEASURING THE CONCENTRATION OF HIGH OR LOW VALUES

Geary's c and Moran's I measure whether similar values are clustered or dispersed. If you want to know whether high or low values cluster, use the General G-statistic, which measures the concentration of values over an area.

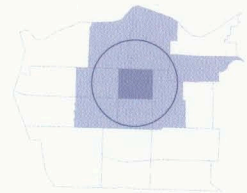
What the G-statistic measures

The G-statistic tells you whether either hot spots (clusters of high values) or cold spots (clusters of low values) exist in a study area. However, you don't know where the concentration of values is—only that across the study area, the high (or low) values tend to occur near each other. An entrepreneur opening a chain of athletic clothing stores in a region would use the G-statistic to see if there are concentrations of people likely to spend money on sporting goods, then start to look at specific locations for the stores.



Median income by ZIP Code. The General G-statistic tells you if there are clusters of ZIP Codes having high values for this attribute.

The G-statistic uses a neighborhood based on a distance you specify. Feature pairs for which the neighboring feature is within the distance of the target feature are assigned a weight of 1; all other pairs are assigned a weight of 0 (see “Defining spatial neighborhoods and weights”).



Calculating the G-statistic

The GIS multiplies the attribute values for the first feature pair, then multiplies this product by the weight—1 if the features are within the distance you specified, or 0 if they aren't. It then does the same for all other pairs of features in the dataset and sums the results. For pairs where the distance is greater than the specified distance, the value ends up being 0 (since the weight is 0). Those pairs count for nothing. Finally, the sum is divided by the sum of the products of all feature pairs in the dataset.

The General G-statistic,
for a distance (d)

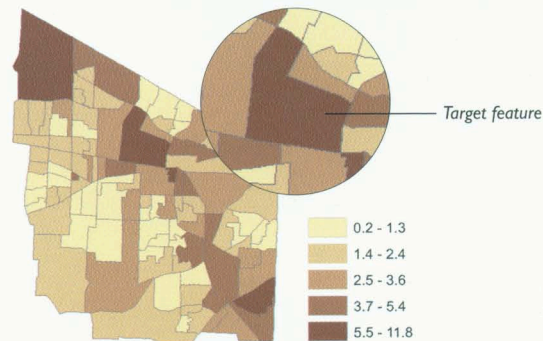
The value of the target feature (x_i) is multiplied by the value of each neighbor (x_j); the product is multiplied by the weight for that pair (w_{ij}), and added to the sum....

$$G(d) = \frac{\sum_i \sum_j w_{ij} (x_i \cdot x_j)}{\sum_i \sum_j (x_i \cdot x_j)}$$

....then the weighted sum is divided by the unweighted sum of the products of all features

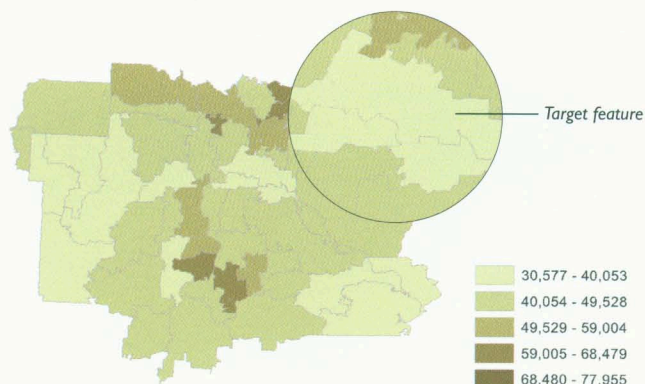
Interpreting the results of the G-statistic

A large value for G indicates that high attribute values are found together, while a small value for G indicates that low values are found together. If you used a distance that included all the features in the dataset, the numerator and denominator would be equal, and G would be 1. In most cases, though, the distance you use will be substantially smaller. Fewer pairs will be included in the numerator, and G will be less than 1. If the pairs within the distance have relatively high values, the numerator, and hence the final value of G, will be larger.



Block groups color coded by number of emergency calls per 100 people. In the zoomed-in area, values surrounding the target feature are relatively high, adding a large number to the sum of values.

If the pairs within the distance have relatively low values, the numerator—and the value of G —will be smaller.



Median income by ZIP Code. The values surrounding the target feature are low, resulting in a small number being added to the sum.

Since the G -statistic is a relative measure, you don't really know what a large or small value means unless you compare it to the expected G -statistic for a random distribution at the distance you specified. You can then tell if the distribution of values is significantly different than a random distribution.

The expected G -statistic at the given distance is what the value of G would be were there no particular concentration of high or low values. It's the ratio of pairs within the distance to the total number of pairs, regardless of value. The GIS calculates this by summing the weights for the pairs and dividing by $n(n-1)$ where n is the number of features in the study area.

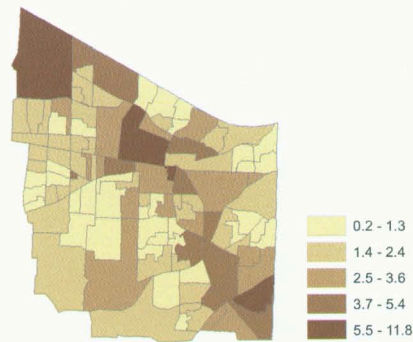
The expected G -statistic at a given distance, for a random distribution

$$G_E(d) = \frac{W}{n(n-1)}$$

The sum of the weights (W)....
...is divided by the number of features (n) times the number of features minus one

Since the weight is 1 for any pair within the distance, W is equal to the number of pairs within the distance. And the denominator is the total number of pairs in the study area.

If the observed G-statistic is larger than the expected G-statistic, then there is a concentration of high values.



This dataset of emergency calls per 100 people has an observed G of 0.07, higher than the expected G of 0.05, indicating that high values are clustered.

If the observed G-statistic is smaller than the expected G-statistic, there is a concentration of low values.



This dataset of median income has an observed G of 0.09, slightly lower than the expected G of 0.10, indicating low values are clustered.

Testing the significance of the G-statistic

Once the GIS has calculated the observed and expected values of G, you can test whether the observed G is significantly different than the expected G (that is, significantly different than a random distribution) at a given confidence level.

The test involves calculating a Z-score. First the GIS calculates the variance for the expected G. The variance represents the average amount that values differ from the mean value for all the features in the study area. Then the expected G is subtracted from the observed G and divided by the square root of the variance (that is, the standard deviation).

$$Z_{G(d)} = \frac{G(d)_O - G(d)_E}{SD_{G(d)}}$$

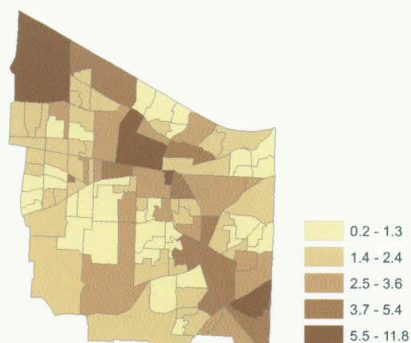
The Z-score for the G-statistic at distance (d)

The expected G value at the distance is subtracted from the observed G...

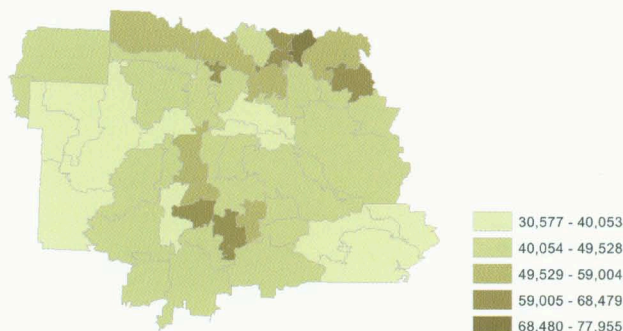
...and the difference divided by the standard deviation for the expected G for that distance

If the observed G is larger than the expected G (that is, high values are clustered), the numerator is positive, and the Z -score is positive. If the observed G is less than the expected G (low values are clustered), the numerator is negative, and the Z -score is negative.

For example, a negative Z -score below the significant value (-1.96 at a confidence level of 95%) indicates that low values tend to be found together and you can be 95% sure the pattern is not due to chance. See the section “Testing statistical significance” for more on the Z -score.



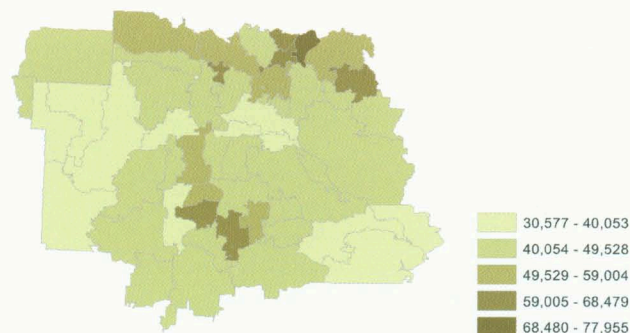
With a G -statistic value higher than expected for a random distribution, and a Z -score of 4.21, the clustering of high values is significant at a confidence level of 0.01 (99%).



With a G -statistic value lower than the expected G for a random distribution (indicating clustering of low values), but a Z -score of -1.26 , the clustering is not statistically significant.

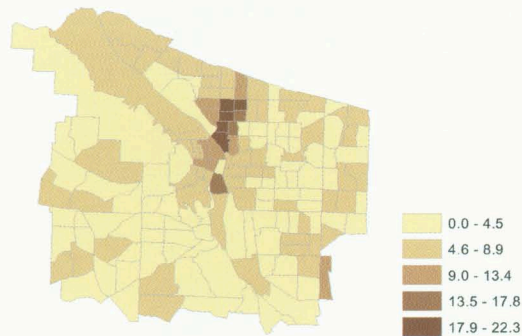
Factors influencing the G -statistic results

The distance you specify when defining the neighborhood will affect the results of the analysis. If, for example, there happen to be more features with high values found within the distance, the statistic will indicate clustering of high values, even though low values may also be clustered.



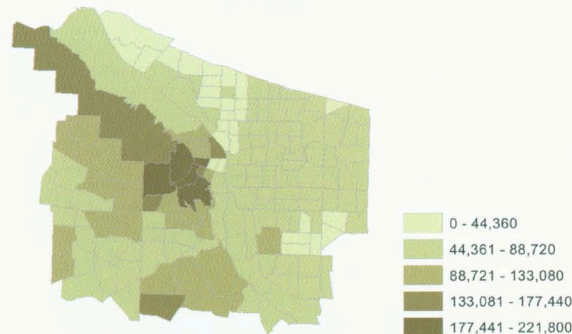
At a distance of a half-mile, the G -statistic indicates clustering of low median income values, with a Z -score of -1.39 , significant at a confidence level of 0.2 or 80%. At a distance of 3 miles, the Z -score is -0.43 , indicating no clustering.

The size of the features you're analyzing can also affect the results. For example, if large areas tend to have low values and smaller areas tend to have high values—and both seem to be equally concentrated—the G-statistic will show that high values are concentrated, since for the small areas there are more pairs within the specified distance.



Census tracts by percentage of vacant parcels. The highest values are in the smaller census tracts, and the G-statistic shows significant clustering of high values.

The results also depend on the range of values for the features. If you have one or a few very high values—relative to the mean value for the dataset—the G-statistic may show that high values are concentrated, even if there are more features with low values near each other. This is because the equation multiplies the feature values. If there are very high values, the numerator will end up being very large.



Median home value by census tract. A few tracts have very high values, causing the G-statistic to indicate significant clustering of high values.